Exercises

Ex. 1. Let X be a weakly stationary process. Show that $\gamma_X(h) = \gamma_X(-h)$.

Correction:

$$var(X_t, X_{t+h}) = Var(X_{t+\ell}, X_{t+\ell+h},)]$$

by definition of the weak stationarity. Then it suffices to take $\ell = -h$.

□

Ex. 2. Let X_t be a real valued weakly stationary process, with autocovariance function γ_X. Let

$$Y_t = X_t - X_{t-1}.$$

Show that Y is weakly stationary and compute its covariance function.

Correction:

$$Cov(Y_t, Y_{t+l}) = E[Y_t Y_{t+k}] - E[Y_t]E[Y_{t+k.}],$$

Now $E[Y_t] = E[X_t] - E[X_{t-1}] = 0$ by stationarity of X. So,

$$\begin{aligned}
Cov(Y_t, Y_{t+l}) &= E[Y_t Y_{t+k}] \\
&= E[(X_t - X_{t-1})(X_{t+k} - X_{t+k-1})] \\
&= E[X_t X_{t+k}] - E[X_{t+k-1}X_t] - E[X_{t+k}X_{t-1}] + E[X_{t+k-1}X_{t-1}]] \\
&= \gamma_X(k) + \mu^2 - \gamma_X(k-1) - \mu^2 - \gamma_X(k+1) - \mu^2 + \gamma_X(k) + \mu^2 \\
&= 2\gamma_X(k) - \gamma_X(k+1) - \gamma_X(k-1)
\end{aligned}$$

where we used $\gamma_X(k) = E[X_t X_{t+k}] - \mu^2$, with $\mu = E[X_t]$.

□

Ex. 3. In this exercise we explore a little more the notion of invertibility.

As explained during the lectures, if $X = F_\alpha Z$ is invertible, there exists β such that $F_\beta X = Z$, and both of these process are causal (that is they only depends on the past). In other words: $F_\beta F_\alpha X = X$, which means $F_\beta = F_\alpha^{-1}$.

If we write these results with polynomials of B:

$$P_\alpha^{-1}(B)P_\alpha(B)X = X.$$

1. If $P_\alpha(B) = 1 - \frac{1}{2}B$, show that $P_\alpha^{-1}(B) = \sum_{n=1}^{\infty} \frac{1}{2^n} B^n$.
2. Show that the AR(1) process $X_t = X_{t-1}/2 + w_t$, where w is a white noise with mean 0 and variance σ^2, it can be written as:

$$X_t = \sum_{i=0}^{\infty} \psi_i w_{t-i},$$

where ψ_i is to be determined.

Ex. 4. Consider the MA(2) process defined as:

$$X_t = w_t + 0.7w_{t-1} + 0.2w_{t-2}$$

where w is a white noise with mean 0 and variance σ^2.

1. Show that X is invertible.
2. Find its mean and autocorrelation functions.

Correction:

1. We need to invert $1+0.7B+0.2B^2$. Its roots are (see pre-school results) $\Delta = 0.7^2 - 4 \cdot 1 \cdot 0.2 = 0.49 - 0.8 = -0.31$, $(-0.7 \pm i\sqrt{0.31})/2$ which have modulus strictly inferior to 1.
2. This process is an MA(2), thus we only need to compute $\gamma(0)$, $\gamma(1)$, and $\gamma(2)$/

$$\begin{aligned}
\gamma(0) &= \sigma^2(1 + 0.7^2 + 0.2^2) \\
\gamma(1) &= \sigma^2(0.7 + 0.7 \cdot 0.2) \\
\gamma(2) &= 0.2\sigma^2
\end{aligned}$$

□

Ex. 5 Consider the AR process defined by:

$$X_t = \alpha X_{t-1} - \alpha X_{t-2} + Z_t, \qquad Z_t \sim (0, \sigma^2).$$

Determine for which values of α the process is causal, and for which it is invertible.

Correction: The processus is always invertible (it is written in an inverted form).
The process writes $\Phi(B)X_t = Z_t$, with $\Phi(B) = 1 - \alpha B + \alpha B^2$. The solutions are then in $\mathbb{C}$:

– if $\alpha^2 - 4\alpha = 0$ (that is $\alpha = 0$ or $\alpha = 4$), $\alpha/2$, in this case the process is not causal.

– If $\alpha^2 - 4\alpha < 0$, that is $0 < \alpha < 4$, the complex roots are $\frac{\alpha \pm i\sqrt{4\alpha - \alpha^2}}{2\alpha}$, after computation we need $0 < \alpha < 1$.

– If $\alpha^2 - 4\alpha > 0$, that is either $\alpha > 4$ and in this case the process is non causal (see above), or $\alpha < 0$ in this case we must also have $\alpha > -2$ so that $|\alpha - \sqrt{\alpha^2 - 4\alpha}| \leq 2$ [...] In the end, $-1/2 < \alpha < 0$.

□

Ex. 6 Let (w_t) be an iid family of normally distributed RVs with mean 0 and variance σ^2. Let $a, b, c \in \mathbb{R}$ be constants. Determine if each of the following processes is stationary. If so, compute its mean and autocovariance function.

1. $X_t = a - bw_5$;
2. $X_t = w_0 \sin(ct)$;
3. $X_t = w_t w_{t-1}$;
4. $X_t = w_0 \cos(ct) - w_1 \sin(ct)$.

Correction:

1. It is weakly stationary as $E[|X_t|^2] \leq a^2 + 2|a||b|E[w_5]^2 + b^2E[w_5^2]$ is finite, its expectation is $E[X_t] = a - E[w_5] = a$, and its autocovariance is $cov(X_t, X_{t+k}) = var(a - bw_5) = b^2\sigma^2$ is independent of both t and k.

2. We have that $E[X_t^2] = \sin(ct)E[w_0^2]$ is finite. The expected value is 0. The autocovariance function is:

$$cov(X_t, X_{t+k}) = E[w_0^2]\sin(ct)\sin(ct + ck) = \frac{\sigma^2}{2}(\cos(ct - c(t+k)) - \cos(ct + c(t+k)))$$

$$= \frac{\sigma^2}{2}(\cos(ck) - 2\cos(2ct + ck)).$$

Which is independent of t if and only if $c = c_k = \pi k$. But if $c = \pi k$ for some k, then for another $k' \neq k$ the covariance will not be independent of k, thus the process is not weakly stationary.

3. It is weakly stationary as it is L^2 ($E[X_t^2] = E[w_t^4]$ is finite). The expectation is null, and the autocovariance is:

$$cov(X_t, X_{t+k}) = E[w_t w_{t-1} w_{t+k} w_{t+k-1}] = \begin{cases} \sigma^4 & h = 0; \\ 0 & \text{otherwise.} \end{cases}$$

4. L^2-ness is trivial. The mean is obviously null. For the autocovariance:

$$\begin{aligned}
Cov(X_t, X_{t+k}) &= E[(w_0\cos(ct) - w_1\sin(ct))(w_0\cos(ct+ck) - w_1\sin(ct+ck))] \\
&= E[w_0^2]cos(ct)cos(ct+ck) + E[w_1^2]sin(ct)sin(ct+ck) \\
&\quad - E[w_0]^2cos(ct)cos(ct+ck) - E[w_1]^2sin(ct)cos(ct+ck). \\
&= \sigma^2(cos(ct)cos(ct+ck) + sin(ct)sin(ct+ck)) \\
&= \sigma^2\left(\frac{1}{2}(cos(ck) + cos(2ct+ck)) + \frac{1}{2}(cos(ck) - cos(2ct+ck))\right) \\
&= \sigma^2 cos(ck)
\end{aligned}$$

which only depends on k.

□

Ex. 7 Let (x_t) be the time serie of the average yearly temperature in New Haven (USA). We assume these observations comfe from a stationary time series (it was before the largest effects of global warming). The observations are given below. They are also available on `R`, in the `datasets` package (10 first obserbations of `nhtemp`).

Year	1912	1913	1914	1915	1916	1917	1918	1919	1920	1921
Temp (°F)	49,9	52,3	49,4	51,1	49,4	47,9	49,8	50,9	49,3	51,9

Explain mathematically what you do and do the computations and plots with `R`.

1. Estimate $E[X_t]$ and explain why you chose that estimator. Compute the autocorrelation function.

2. Draw a correlogram at the first 3 lags.

3. Under the null hypothesis that (X_t) is a white noise, write your approximate distribution for the estimated autocorrelation coefficients r_1, r_2, r_3.

4. Perform tests to asses whether the process is a white noise using the Binomial distribution and the Ljung-Box test independently.

5. compare those results when using the full set of observations.

Correction:

1. Weak law of large numbers, empirical mean, for the other we use empirical variance.

$$E[X_t] = 50.19, c_0 = 1.623, c_1 = -0.345, c_2 = 0.093 \text{ and } c_3 = 0.033. r_0 = 1,$$

$$r_1 = -0.212, r_2 = 0.057 \text{ and } r_3 = 0.021$$

2. use R.

3. The approximate distribution only depends on the number of observations $r_k \sim \mathcal{N}(0, 1/10)$.

4. If the process were white noise we would expect around 95% of the r_k to lie between $\pm 1.96/\sqrt{10} = \pm 0.6198$ (for $k \in \{1, 2, 3\}$). Using the binomial distribution, we would therefore expect the number of r_k lying outside these limits to be distributed as Binomial$(3, 0.05)$. This distribution has mean 0.15 and variance 0.1425. All the r_k are within the limits, which is clearly not unusual under the assumed binomial distribution. We therefore have insufficient evidence to reject the null hypothesis.

   The Ljung-Box statistic is:

$$LB = n(n+2) \sum_{k=1}^{m} \left[\frac{r_k^2}{n-k} \right] \simeq 0.65$$

   we compare it to the $\chi_3^2(0.95) = 7.8$, there's insufficient evidence to reject the null hypothesis.

5. Check with `R`, we get $LB = 19.8$ which is sufficient evidence to reject null hypothesis.

□

Ex. 8 For each of the Figures 1, 2, 3 and 4, observe the correlogram and the PACF. State what model you would choose and why. Give the order of the models and list any approximate parameter that might be estimated from the graphs.

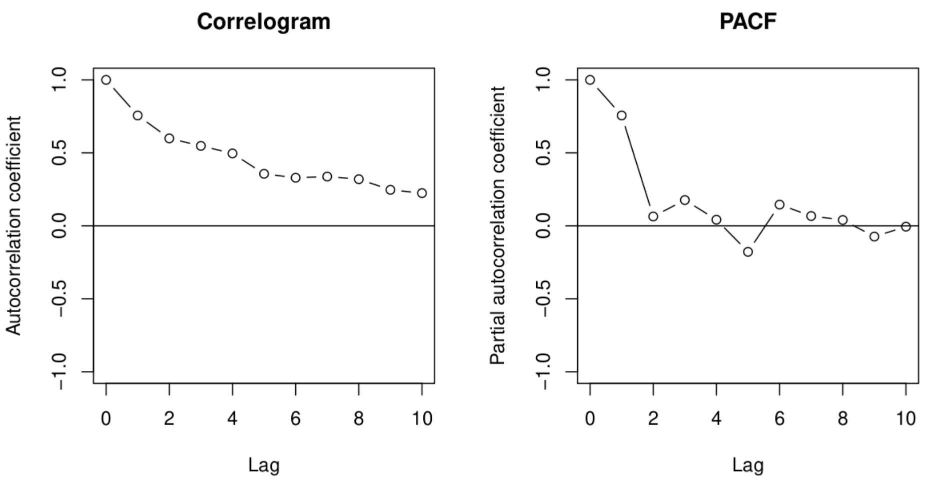

Figure 1: (a)

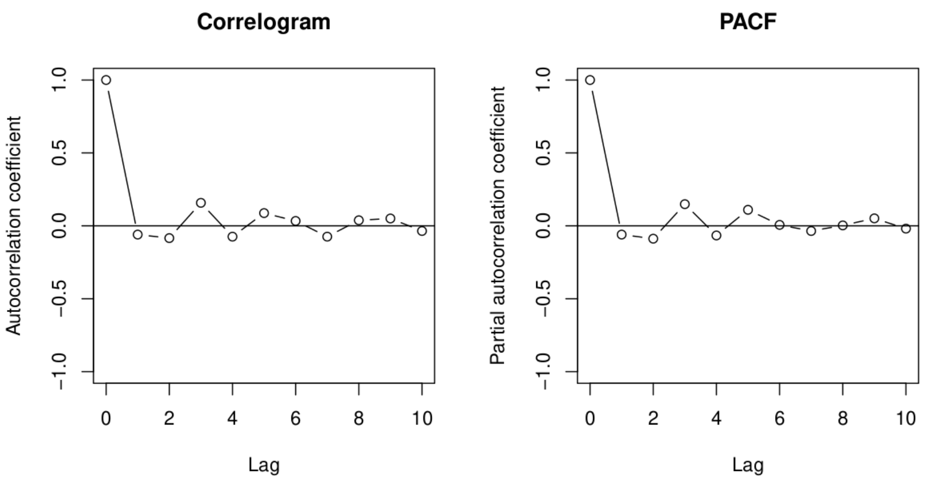

Figure 2: (b)

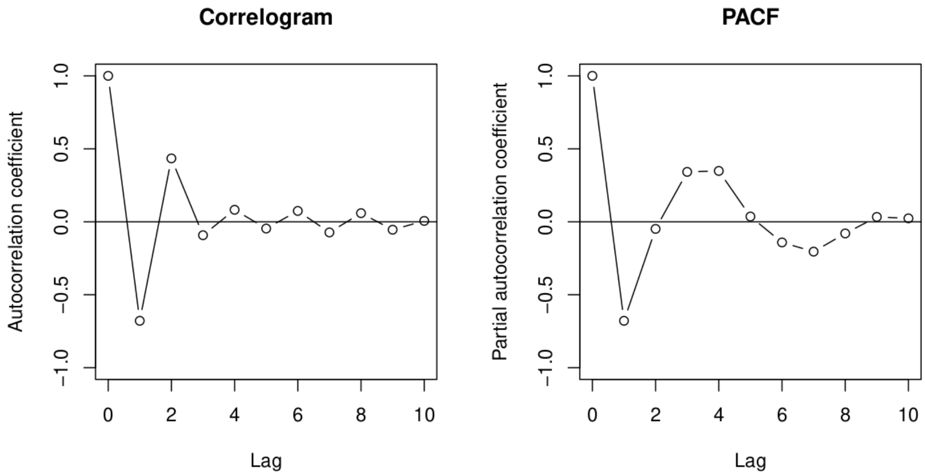

Figure 3: (c)

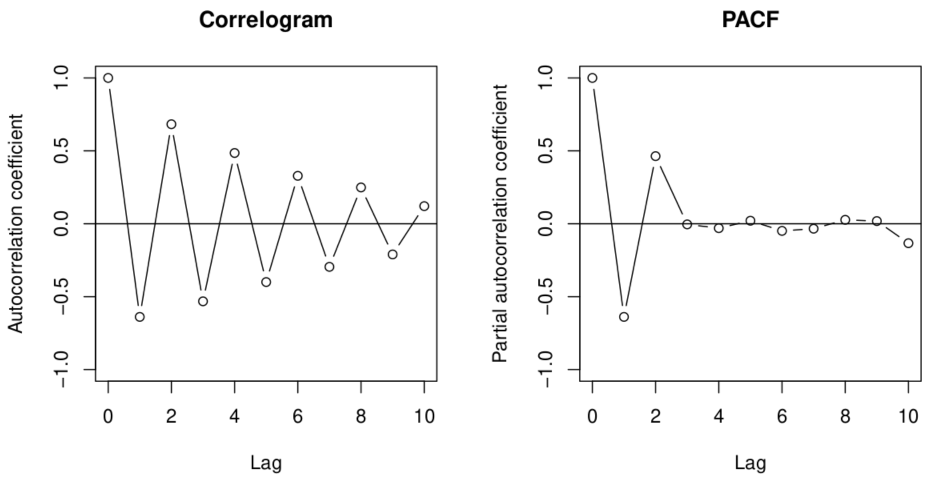

Figure 4: (d)

Correction:

1. AR(1), as the PACF goes to 0 after that. The first autocorrelation parameter can be approximated by $0,8$.
2. White noise: all the autocorrelation coefficient (except at lag 0) are approximately null. The covariance of the model is approximately 1.
3. MA(2) as the correlogram goes to 0 after lag 2.
4. AR(2) as PACF is close to 0 after lag 2. An estimate of the second autocorrelation parameter is $0,5$.

$\square$

Ex. 9 We define the seasonal difference $\Delta_d^k = (1 - B^d)^k$, and for short $\Delta_: := \Delta_1^1$; $\Delta_d := \Delta_d^1$; $\Delta^k = \Delta_1^k$. Show that:

1. if (Y_t) is a weakly stationary process, then for arbitrary $k \in \mathbb{N}^\star$ the process $(\Delta^k Y_t)$ is again weakly stationary.
2. If (Y_t) is a weakly stationary process then for arbitrary $d \in \mathbb{N}^\star$ the process $(\Delta_d Y_t)$ is again weakly stationary.

Correction:

1. Let (Y_t) be a weakly stationary process. Applying the difference operator at order k leads to:

$$\Delta^k Y_t = (1-B)^k Y_t = \sum_{j=0}^{k} \binom{k}{j} 1^{k-j} (-B)^j Y_t =: \sum_{j=0}^{k} (-1)^j a_j Y_{t-j}.$$

Therefore, we have L_2-ity of the process. The expected value is equal to:

$$E[\Delta^k Y_t] = E[Y_0] \sum_{j=0}^{k} (-1)^j a_j.$$

And the autocovariance:

$$\begin{aligned}
cov(\Delta^k Y_t, \Delta^k Y_{t+h}) &= E[\Delta^k Y_t \Delta^k Y_{t+h}] - Cst \\
&= E\left[\sum_{j=0}^{k} (-1)^j a_j Y_{t-j} \sum_{i=0}^{k} (-1)^i a_i Y_{t-i+h}\right] - Cst \\
&= \sum_{j=0}^{k} \sum_{i=0}^{k} (-1)^{i+j} a_j a_i E[Y_{t+h} Y_t] - Cst \\
&= \sum_{j=0}^{k} \sum_{i=0}^{k} (-1)^{i+j} a_j a_i (\gamma_Y(h-j+i) + E[Y_0]^2) - Cst.
\end{aligned}$$

This last expression is independent of t so the process is weakly stationary.

2. This is similar to the last question with a general d. The autocovariance writes:

$$\gamma_{\Delta_d Y}(h) = \gamma_Y(h) - \gamma_Y(h+d) - \gamma_Y(h-d) + \gamma_Y(h).$$

□

Ex. 10 Let $x_1, \dots, x_n$ be n observations of an AR(p) process associated with white noise of parameters $(0, \sigma^2)$. The coefficients $\hat{\alpha_1}, \dots, \hat{\alpha_n}$ have been estimated by minimizing

$$S = \frac{1}{n} \sum_{t=p+1}^{n} \left(x_t - \sum_{i=1}^{p} \alpha_i x_{t-i} \right)^2 .$$

1. Assume that the observations come from an AR(1), show that the least square estimators can be obtained by minimising:

$$S = \frac{1}{n} \sum_{t=2}^{n} (x_t - \alpha x_{t-1})^2 .$$

2. Solve the previous minimising problem.
3. Compare your estimators to the estimation of the autocorrelation coefficient given in the lecture notes.
4. Assume you observe the following time series:

$$1,658; -3,194; -1,402; -1,426; 0,609; 0,575; -0,241; 0,030; 0.171; -0,260.$$

Fit the AR(1) model with zero mean given by: $X_t = \alpha_1 X_{t-1} + Z_t$, $(Z_t) \sim (0, \sigma^2)$.

Correction:

1. Trivial, replace p by 1.
2. We derive the expression of S w.r.t. α:

$$\frac{\mathrm{d}S}{\mathrm{d}\alpha} = -\frac{2}{n} \sum_{2}^{n} x_{t-1}(x_t - \alpha x_{t-1}).$$

Setting this to 0 and solving for α we get,

$$\begin{aligned} 0 &= \sum_{t=2}^{n} x_{t-1}(x_t - \hat{\alpha} x_{t-1}) \\ &= \sum_{t=2}^{n} x_{t-1} x_t - \hat{\alpha} \sum_{t=2}^{n} x_{t-1}^2 \\ \Leftrightarrow \alpha &= \frac{\sum_{t=2}^{n} x_{t-1} x_t}{\sum_{t=2}^{n} x_{t-1}^2}. \end{aligned}$$

3. This estimate is almost the same as the r_1 estimate given in the notes when the mean of the process is assumed null.

4. Use `R`, the estimate is 0.624.

□

Ex. 11 Consider the ARMA$(2,2)$ model defined by:

$$X_t - \frac{13}{20}X_{t-1} - \frac{3}{25}X_{t-2} = w_t - \frac{2}{10}w_{t-1} + \frac{1}{10}w_{t-2}$$

with $(w_t) \sim (0, \sigma^2)$. Determine if the process is regular.

Correction: With the notations of the notes, $\Phi(B) = 1 - \frac{13}{20}B - \frac{3}{25}B^2$ and $\Theta(B) = 1 - \frac{2}{10}B + \frac{1}{10}B^2$. The roots of Φ are:

$$\frac{0.65 \pm 0.95}{-0.24}.$$

And the roots of Θ are:

$$1 \pm 5i\sqrt{0.36}.$$

The polynoms don't have roots in common and the roots all lie outside of the unit circle, therefore the ARMA is regular.

□

Ex. 12 Consider the ARMA$(2,2)$ model

$$X_t + 0.3X_{t-1} - 0.1X_{t-2} = w_t + 0.1w_{t-1} - 0.2w_{t-2}$$

with $(w_t) \sim (0, \sigma^2)$.

1. Write the model under th form $\Phi(B)X_t = \Theta(B)w_t$, and show that the model is not regular.

2. Show that the process defined by:

$$Y_t - 0.2Y_{t-1} = w_t - 0.4w_{t-1}$$

with $(w_t) \sim (0, \sigma^2)$, is such that $Y_t + 0,5Y_{t-1}$ is equivalent to the process defined at the last question.

3. Show that the ARMA model of the previous question can be obtained from the equation of the first question.

Correction:

1. We can factorise Φ into $(1+0,5B)(1-0,2B)$, and Θ into $(1+0,5B)(1-0,4B)$. Thus the root -2 is common to both polynoms.

2.
$$Y_t + 0,5Y_{t-1} = 0,2Y_{t-1} + w_t - 0,4w_{t-1} + 0.1Y_{t-2} + 0.5w_{t-1} - 0.2w_{t-2};$$
$$Y_t = -0,3Y_{t-1} + 0,1Y_{t-2} + w_t + 0,1w_{t-1} - 0,2w_{t-2}.$$

   Which is the process given in the previous question.

3. We can cancel out the common factor in the writing $\Phi(B)X_t = \Theta(B)w_t$, getting the process defined in the second question (just check the factorized form...)

□

Ex. 13 Consider the ARMA$(1,2)$ process defined by:

$$X_t = \alpha X_{t-1} + w_t + \beta w_{t-1} + \beta^2 w_{t-2}$$

where $(w_t) \sim (0, \sigma^2)$ and $|\alpha|, |\beta| < 1$.

1. Show that this process is causal and invertible.

2. Show that the process can be written under the form:

$$X_t = \sum_{i=0}^{\infty} \psi_i w_{t-i}$$

   where ψ is to be defined.

3. Find an expression for $Var(X_t)$.

Correction:

1. We have $\Phi(B)X_t = \Theta(B)w_t$, with

$$\Phi(B) = (1 - \alpha B)$$
$$\Theta(B) = (1 + \beta B + \beta^2 B).$$

   The root of Φ is $1/\alpha$, as $|\alpha| < 1$ the process is causal. The roots of Θ are:

$$\frac{-\beta \pm \sqrt{\beta^2 - 4\beta^2}}{2\beta^2} = -\frac{1}{2\beta} \pm \frac{i\sqrt{3}}{2\beta}.$$

   These complex roots have modulus $1/|\beta|$, and thus the process is invertible.

2. We use Wold decomposition, using that $(1-\alpha B)^{-1} = 1 + \alpha B + \alpha^2 B + \dots$:

$$X_t = (1 + (\alpha+\beta)B + (\alpha^2+\alpha\beta+\beta^2)B^2 + \alpha(\alpha^2+\alpha\beta+\beta^2)B^3 + \alpha^2(\alpha^2+\alpha\beta+\beta^2)B^4 + \dots)w_t.$$

   That is the form required with $\psi_0 = 1, \psi_1 = (\alpha+\beta)$ and $\psi_i = \alpha^{i-2}(\alpha^2+\alpha\beta+\beta^2)$ for $i = 2, 3, \dots$.

3. We use Wold decomposition:

$$\begin{aligned} Var(X_t) &= \sum_{i=0}^{\infty} \psi_i Var(w_{t-i}) \\ &= (1 + (\alpha+\beta)^2) + (\alpha^2+\alpha\beta+\beta^2)^2(1+\alpha^2+\alpha^4+\dots)\sigma^2. \end{aligned}$$

□

Ex. 14 Consider the process

$$X_t = X_{t-1} + 0,2X_{t-1} - 0,2X_{t-2} + w_t - 0,5w_{t-1}$$

with $(w_t) \sim (0, \sigma^2)$.

1. Classify the model as an ARIMA(p, d, q) that is determine the orders.
2. Show that the model can be written $\Phi(B)X_t = \Theta(B)w_t$.
3. Find the roots of Φ and Θ.
4. Show that w_t can be written:

$$w_t = \sum_{i=0}^{\infty} \psi_i X_{t-i}$$

   and evaluate ψ_0, ψ_1, ψ_2 and ψ_3.

Correction:

1. Using the difference operator:

$$\Delta X_t = 0,2\Delta X_{t-1} + w_t - 0,5w_{t-1}.$$

   Therefore, it is ARIMA$(1, 1, 1)$.

2. $\Delta X_t = (1-B)X_t$, thus

$$\begin{aligned} (1-B)X_t - 0,2(1-B)X_{t-1} &= w_t - 0,5w_{t-1} \\ (1-0,2B)(1-B)X_t &= (1-0,5B)w_t. \end{aligned}$$

   which is the required form.

3. The roots are written above.
4. As the process is invertible, we can write:

$$\begin{aligned} w_t &= (1-B)(1-0,2B)(1+0,5B+0,5^2B^2+\dots)X_t \\ &= (1+(0,5-1,2)B+(0,5^2-0,6+0,2)B^2+(0,5^3-0,3+0,1)B^3+\dots)X_t. \end{aligned}$$

Thus, $\psi_0 = 1$, $\psi_1 = -0,7$, $\psi_2 = -0,15$, $\psi_3 = -0,075$.

□

Ex. 15 Consider the moving average model

$$X_t = w_t + \theta w_{t-1},$$

for $|\theta| < 1$, where $\{w_t\} \sim (0, \sigma^2)$.

1. Using the Box-Jenkins forecasting method, show that $X^n_{n+1} = \theta z_n$, where X^n_{n+1} is the forecast of time $n+1$ from time n and z_n is the observed error at time n.
2. Show that $X^n_{n+h} = 0$ for $h = 2, 3 \dots$.
3. Show that the variance of the h-step ahead forecast error ($e_n(h)$) is σ^2 for $h = 1$ and $(1+\theta^2)\sigma^2$ for $h \geq 2$, if θ is known.

Correction:

1. To use the Box-Jenkins forecasting method, set $w_{n+1} = 0$, so that

$$X^n_{n+1} = \theta z_n.$$

2. The h-steps ahead forecast is given by

$$X^n_{n+h} = w_{n+h} + \theta w_{n+h-1}.$$

Future values of w_t should be set to zero, so if $h > 1$, $w_{n+h} = 0$ and $w_{n+h-1} = 0$. Hence, $X^n_{n+h} = 0$.

3. The error $e_n(h)$ is given by

$$e_n(h) = X^n_{n+h} - X_{n+h}.$$

When $h = 1$, the variance of $e_n(h)$ is

$$\begin{aligned} \text{Var}[e_n(1)] &= \text{Var}[\theta z_n - w_{n+1} - \theta z_n] \\ &= \text{Var}[w_{n+1}] = \sigma^2. \end{aligned}$$

When $h > 1$, the variance is

$$\begin{aligned} \text{Var}[e_n(h)] &= \text{Var}[0 - w_{n+h} - \theta w_{n+h-1}] \\ &= \text{Var}[w_{n+1}] + \theta^2 \text{Var}[w_{n+h-1}] = \sigma^2(1+\theta^2). \end{aligned}$$

□

Ex. 16 (May 2012 Q1 on exam paper).
Consider the time series model

$$X_t = X_{t-1} + w_t - \theta w_{t-1},$$

where $0 < \theta < 1$ and $\{w_t\} \sim (0, \sigma^2)$.

1. Classify this process as an ARIMA(p, d, q) model, where you should specify p, d and q.
2. Write the model in the form
$$\Phi(B)X_t = \theta(B)w_t,$$
where $\Phi(B)$ and $\theta(B)$ are polynomials in B. Find the roots of these polynomials.
3. Show that we may write
$$w_t = \pi(B)X_t,$$
where $\pi(B) = 1 + \pi_1 B + \pi_2 B^2 + \ldots$, and give an expression for π_i.
4. Let X^n_{n+h} denote the Box-Jenkins forecast at time n for X_{n+h}. Write down an expression for X^n_{n+1} in terms of θ and $X_n, X_{n-1}, \ldots$.
5. What is the name of this forecasting method?
6. In practice, X_t may only be available for $t = n, n-1, \ldots 1$. Modify the formula of (d) so that the sum of the coefficients of $X_n, X_{n-1}, \ldots X_n$ is equal to one.

Correction:

1. Let $W_t = X_t - X_{t-1} = \Delta X_t$. We have
$$W_t = \Delta X_t = (1 - \theta B)w_t,$$
so X_t is ARIMA(0,1,1).
2. We have
$$\Delta X_t = (1 - B)X_t = (1 - \theta B)w_t,$$
so letting $\Phi(B) = (1 - B)$ and $\theta(B) = (1 - \theta B)$, we can write $\Phi(B)X_t = \theta(B)w_t$. The root of $\Phi(B)$ is 1. The root of $\theta(B)$ is $1/\theta$.
3. We know that $0 < \theta < 1$, so the process is invertible. Hence we may write $w_t = \pi(B)X_t$, where
$$\begin{aligned}
\pi(B) &= \theta(B)^{-1}\Phi(B) \\
&= \frac{(1 - B)}{(1 - \theta B)} = (1 - B)(1 + \theta B + \theta^2 B^2 + \ldots) \\
&= 1 + \theta B + \theta^2 B^2 + \ldots - B - \theta B - \ldots \\
&= 1 - (1 - \theta)B - (1 - \theta)\theta B^2 - (1 - \theta)\theta^2 B^3 + \ldots \\
&= 1 + \pi_1 B + \pi_2 B + \ldots
\end{aligned}$$
where $\pi_i = -(1 - \theta)\theta^{i-1}$, for $i = 1, 2, 3 \ldots$.

4. We have that $X_{n+1} = X_n + w_{n+1} - \theta w_n$ and so $X_{n+1}^n = X_n - \theta w_n$. Using part (c), we can write

$$\begin{aligned} X_{n+1}^n &= X_n - \theta w_n \\ &= X_n - \theta(X_n - (1-\theta)X_{n-1} - (1-\theta)\theta X_{n-2} - (1-\theta)\theta^2 X_{n-3} + \ldots) \\ &= (1-\theta)X_n + \theta(1-\theta)X_{n-1} + \theta^2(1-\theta)X_{n-2} + \ldots \end{aligned}$$

5. This is also known as exponential smoothing.

6. There were two methods given in the lecture notes for adjusting the weights in the above formula. There are

   (a) increase the weight of X_1 to w, such that

   $$(1-\theta) + (1-\theta)\theta + (1-\theta)\theta^2 + \ldots + (1-\theta)\theta^{n-2} + w = 1,$$

   so that
   $$w = 1 - (1-\theta)\frac{(1-\theta^{n-1})}{(1-\theta)} = \theta^{n-1}.$$

   (b) multiply the weights for $X_n, X_{n-1}, \ldots X_1$ by a factor f, such that

   $$f[(1-\theta) + (1-\theta)\theta + \ldots + (1-\theta)\theta^{n-1}] = 1$$

   so that
   $$f = \left[(1-\theta)\left(\frac{1-\theta^n}{1-\theta}\right)\right]^{-1} = \frac{1}{1-\theta^n}.$$

□

Ex. 17 Let X_t denote the daily income ($000's) of a store in Michigan, USA. The owner is interested in forecasting the daily income for tomorrow. The income of the store is thought to follow the process
$$\Delta^2 X_t = w_t + 0.748 w_{t-1},$$
where $\{w_t\} \sim (0, \sigma^2)$.

1. Describe the model using the ARIMA(p,d,q) notation.
2. Find a formula for X_{n+1}^n, using the Box-Jenkins method, in terms of X_n, X_{n-1} and X_n^{n-1}.

Correction:

1. The process is ARIMA(0, 2, 1).
2. We have
   $$X_{n+1} = 2X_n - X_{n-1} + w_{n+1} + 0.748 w_n$$
   and so
   $$X_{n+1}^n = 2X_n - X_{n-1} + 0.748 w_n.$$

But we know that $w_n = X_n - X_n^{n-1}$, so

$$\begin{aligned} X_{n+1}^n &= 2X_n - X_{n-1} + 0.748(X_n - X_n^{n-1}) \\ &= 2.748X_n - X_{n-1} - 0.748X_n^{n-1} \end{aligned}$$

Therefore, the sales forecast for tomorrow may be expressed in terms of today's sales, yesterday's sales and yesterday's forecast for today's sales.

□

Ex. 18 Let the time series $\{X_t\}$ denote the average height (in feet) of Lake Huron at time t. Observations of $\{X_t\}$ for each of the years from 1963 to 1972 are given in the following table

Year	1963	1964	1965	1966	1967	1968	1969	1970	1971	1972
Height	576.89	575.96	576.80	577.68	578.38	578.52	579.74	579.31	579.89	579.96

1. Take the forecast for 1963 (made in 1962) to be 579 (so that $X_{1963}^{1962} = 579$). Use exponential smoothing (the recursive formula) to obtain a forecast for the level of Lake Huron in the year 1973, setting $\theta = 0.5$.
2. Now use Holt's method to estimate the level of Lake Huron in the year 1973, taking the forecast for 1964 made in 1963 to be 579.5 (so that $X_{1964}^{1963} = 579.5$). Assume that $\alpha = 0.3$ and $\gamma = 0.5$.

Correction:

1. The formula

$$X_{n+1}^n = 0.5X_n + 0.5X_n^{n-1}$$

can be used to obtain forecasts for each year. Starting with the given value of 579, we get

Year	Height	Forecast
1963	576.89	579
1964	575.96	577.945
1965	576.8	576.9525
1966	577.68	576.87625
1967	578.38	577.278125
1968	578.52	577.8290625
1969	579.74	578.1745313
1970	579.31	578.9572656
1971	579.89	579.1336328
1972	579.96	579.5118164
1973		579.7359082

The forecast obtained for 1973 is 579.7359.

2. To use Holt's forecasting method, we use the formula

$$X_{n+1}^n = \alpha(1+\gamma)X_n - \alpha X_{n-1} + (2 - \alpha - \alpha\gamma)X_n^{n-1} - (1-\alpha)X_{n-1}^{n-2}$$

with $\alpha = 0.3$ and $\gamma = 0.5$. We obtain forecasts

Year	Height	Forecast
1963	576.89	579
1964	575.96	579.5
1965	576.8	579.04
1966	577.68	578.634
1967	578.38	578.4707
1968	578.52	578.552785
1969	579.74	578.6473268
1970	579.31	579.243407
1971	579.89	579.5416521
1972	579.96	579.9766758
1973		580.2996911

The forecast for 1973 is 580.2996911.

□

Ex. 19 (May 2012 Q5 on exam paper)

1. Let $\{x_t\}$ denote a set of observations from a quarterly time series which is thought to be subject to seasonal variations. Describe briefly how to estimate the quarterly seasonal variations by first calculating a suitable adjusted average, y_t.

The following table refers to a quarterly index of vegetable prices in the UK from the first quarter of 1955 to the first quarter of 1959.

Date	x_t	y_t	$x_t - y_t$
Q1 1955	333.7		
Q2 1955	323.9		
Q3 1955	312.8	318.8375	-6.0375
Q4 1955	310.2	319.9000	-9.7000
Q1 1956	323.2	320.7125	2.4875
Q2 1956	342.9	319.1000	23.8000
Q3 1956	300.3	316.6875	-16.3875
Q4 1956	309.8	307.2000	2.6000
Q1 1957	304.3	299.0750	5.2250
Q2 1957	285.9	296.6875	-10.7875
Q3 1957	292.3	296.3250	-4.0250
Q4 1957	298.7	303.6250	-4.9250
Q1 1958	312.5	310.3000	2.2000
Q2 1958	336.1	313.1625	22.9375
Q3 1958	295.5	314.0750	-18.5750
Q4 1958	318.4		
Q1 1959	300.1		

2. Estimate the seasonal variations in quarters 1 to 4 respectively.

3. Use the estimated variations to find the quarter in which vegetable prices tended to be the highest.

4. Find the seasonally-adjusted series.

Correction:

1. To estimate the quarterly seasonal variations, the following steps should be taken

   – Let x_t (the raw observation) be equal to $u_t + e'(t) + s(t)$, for $(1 \leq t \leq n)$, where u_t is the trend, $e'(t)$ is the random error and s_t is the seasonal variation.

   – The raw data can then be smoothed using a suitable adjusted average (for quarterly data, this could be using the formula
   $$y_t = \frac{1}{8}x_{t-2} + \frac{1}{4}(x_{t-1} + x_t + x_{t+1}) + \frac{1}{8}x_{t+2}.$$

   – The error $e_t = x_t - y_t$ can then be calculated, where e_t is the difference between the raw observations and the smoothed series.

   – The seasonal variation for each quarter can then be calculated as the average of the e_t over all observations in each quarter (e.g. the seasonal variation for quarter one is the average of the e_t for all t in quarter one.

   – The seasonal variations should then be adjusted so that they sum to zero.

2. The initial estimates of the seasonal variations (i.e. before adjusting so that they sum to one) are given by
   $$\begin{aligned}
   s_*^I &= \frac{1}{3}(2.4875 + 5.225 + 2.2) = 3.304 \\
   s_*^{II} &= \frac{1}{3}(23.8 - 10.7875 + 22.9375) = 11.983 \\
   s_*^{III} &= \frac{1}{4}(-6.0375 - 16.3875 - 4.025 - 18.575) = -11.256 \\
   s_*^{IV} &= \frac{1}{3}(-9.7 + 2.6 - 4.925) = -4.008
   \end{aligned}$$
   The adjustment to be made is
   $$\text{adj} = \frac{1}{4}(3.304 + 11.983 - 11.256 - 4.008) = 0.006$$
   Subtracting adj from the initial estimates of the seasonal variation, we get seasonal variations of
   $$\begin{aligned}
   s^I &= 3.298 \\
   s^{II} &= 11.977 \\
   s^{III} &= -11.262 \\
   s^{IV} &= -4.014
   \end{aligned}$$

3. The second quarter has the highest seasonal variation.

4. To get the adjusted series, we subtract 3.298 from Q1 observations, 11.977 from Q2 observations, etc., getting

Date	Adjusted series
Q1 1955	330.402
Q2 1955	311.922
Q3 1955	324.062
Q4 1955	314.214
Q1 1956	319.902
Q2 1956	330.922
Q3 1956	311.562
Q4 1956	313.814
Q1 1957	301.002
Q2 1957	273.922
Q3 1957	303.562
Q4 1957	302.714
Q1 1958	309.202
Q2 1958	324.122
Q3 1958	306.762
Q4 1958	322.414
Q1 1959	296.802

□

Ex. 20 (May 2011 exam question) Woolhouse's adjusted-average formula is central, with coefficients

$$\frac{1}{125}\left(-3,-2,0,3,7,21,24,25,24,21,7,3,0,-2,-3\right).$$

1. State the length of this filter.
2. Show that this filter is exact on cubics.

Correction:

1. The length of the filter is 15.
2. For the filter to be exact on cubics we must have that

   (a) $\sum_{j=-7}^{7} K_j = 1$, and

   (b) $\sum_{j=-7}^{7} j^2 K_j = 0$.

   We find that

$$\sum_{j=-7}^{7} K_j = \frac{1}{125}\left(-3-2+3+7+21+24+25+24+21+7+3-2-3\right) = 1,$$

and

$$\sum_{j=-7}^{7} j^2 K_j = \frac{1}{125}(-3(49) - 2(36) + 3(16) + 7(9) + 21(4) + 24(1)$$
$$+24(1) + 21(4) + 7(9) + 3(16) - 2(36) - 3(49)) = 0,$$

so the conditions hold and the filter is exact on cubics.

$\square$

Ex. 21 (From Chatfield (2004)) Suppose we have a seasonal series of monthly observations $\{X_t\}$, for which the seasonal factor at time t is denoted by $\{S_t\}$. Further suppose that the seasonal pattern is constant through time so that $S_t = S_{t-12}$ for all t. Denote a weakly stationary series of random observations by $\{\epsilon_t\}$.

1. Consider the model

$$X_t = a + bt + S_t + \epsilon_t,$$

having a linear trend and additive seasonality. Show that the difference operator $(1 - B^{12})$ acts on $\{X_t\}$ to produce a weakly stationary series.

2. Now consider the model

$$X_t = (a + bt)S_t + \epsilon_t.$$

This model is said to have **multiplicative** seasonality (as well as a linear trend). Does the difference operator $(1 - B^{12})$ transform $\{X_t\}$ to stationarity? If not, find a difference operator that does.

3. Consider the linear filter

$$\sum_{j=\alpha}^{\beta} K_j X_{t+j},$$

applied to the time series $\{X_t\}$. Show that if

(a) $\sum_{j=\alpha}^{\beta} K_j = 1$, and

(b) $\sum_{j=\alpha}^{\beta} j^i K_j = 0$, for all $i \leq k$,

then the filter does not distort trends which are polynomials of degree k (so is exact on polynomials of degree k).

4. Consider the symmetric linear filter

$$\sum_{j=-3}^{3} K_j X_{t+j},$$

applied to the time series $\{X_t\}$.

(a) Find expressions for K_1, K_2 and K_3 in terms of K_0 such that the filter is exact on polynomials of degree five (you will need to use the theorem you proved in question 3).

(b) Hence, find a symmetric linear filter that is exact on polynomials of degree five and has coefficients which minimise the expression

$$R_0^2 = \sum_{j=-3}^{3} (K_j^2).$$

Correction:

1. Applying the given difference operator to the series, we get

$$\begin{aligned}(1 - B^{12})X_t =&(1 - B^{12})(a + bt + S_t + \epsilon_t)\\ =&a + bt + S_t + \epsilon_t - a - b(t - 12) - S_{t-12} - \epsilon_{t-12}\\ =&12b + \epsilon_t - \epsilon_{t-12},\end{aligned}$$

which is a weakly stationary series, with constant.

2. Applying the difference operator to the series with multiplicative seasonality, we get

$$\begin{aligned}(1 - B^{12})X_t =&(1 - B^{12})((a + bt)S_t + \epsilon_t)\\ =&aS_t + btS_t + \epsilon_t - aS_{t-12} - b(t - 12)S_{t-12} - \epsilon_{t-12}\\ =&12bS_{t-12} - \epsilon_{t-12} + \epsilon_t.\end{aligned}$$

This is not a stationary series, as there is still seasonality. However, applying the same difference operator again, we get

$$\begin{aligned}(1 - B^{12})(12bS_{t-12} - \epsilon_{t-12} + \epsilon_t) =&12bS_{t-12} - \epsilon_{t-12} + \epsilon_t - 12bS_{t-24} + \epsilon_{t-24} - \epsilon_{t-12}\\ =&\epsilon_t - 2\epsilon_{t-12} + \epsilon_{t-24},\end{aligned}$$

which is weakly stationary. Therefore, the difference operator $(1 - B^{12})^2$ transforms the series with multiplicative seasonality to a stationary series.

3. Assume that the trend u_t is the k-th order polynomial $u_t = a_0 + a_1 t + \ldots a_k t^k$. Then for there

to be no distortion of this trend, we require

$$
\begin{aligned}
u_t' =& u_t = a_0 + \ldots a_k t^k = \sum_{j=\alpha}^{\beta} K_j u_{t+j} \\
=& \sum_{j=\alpha}^{\beta} K_j (a_0 + a_1(t+j) + \ldots a_k (t+j)^k \\
=& a_0 (\sum_{j=\alpha}^{\beta} K_j) + a_1 (\sum_{j=\alpha}^{\beta} K_j (t+j)) + \ldots a_k (\sum_{j=\alpha}^{\beta} K_j (t+j)^k) \\
=& a_0 (\sum_{j=\alpha}^{\beta} K_j) + a_1 (t \sum_{j=\alpha}^{\beta} K_j + \sum_{j=\alpha}^{\beta} K_j j) + \ldots \\
& \ldots + a_k (t^k \sum_{j=\alpha}^{\beta} K_j + k t^{k-1} \sum_{j=\alpha}^{\beta} K_j j + \ldots + \sum_{j=\alpha}^{\beta} K_j j^k) \\
=& a_0 + a_1 t + \ldots a_k t^k,
\end{aligned}
$$

where this last line holds because of the conditions $\sum_{j=\alpha}^{\beta} K_j j^i = 0$ and $\sum_{j=\alpha}^{\beta} K_j = 1$.

4. (a) Using the theorem in question 3, in order for the filter to be exact on polynomials of degree 5 we must have that
   i. $\sum_{j=-3}^{3} K_j = 1$, and
   ii. $\sum_{j=-3}^{3} j^i K_j = 0$, for $i = 1, 2, 3, 4, 5$.

   This latter condition automatically holds for $i = 1$ and $i = 3$ as the filter is symmetric (this also means that $K_{-1} = K_1, K_{-2} = K_2$ and $K_{-3} = K_3$). Using the remaining conditions, we have that

$$\sum_{j=-3}^{3} K_j = K_0 + 2(K_1 + K_2 + K_3) = 1 \tag{1}$$

$$\sum_{j=-3}^{3} j^2 K_j = 2K_1 + 8K_2 + 18K_3 = 0 \tag{2}$$

$$\sum_{j=-3}^{3} j^4 K_j = 2K_1 + 32K_2 + 162K_3 = 0. \tag{3}$$

   By (2) and (3) we have that $24K_2 + 144K_3 = 0$, so that $K_2 = -6K_3$. Thus $2K_1 - 48K_3 + 18K_3 = 0$, so $K_1 - 15K_3 = 0$ and $K_1 = 15K_3$.
   Using (1), we have $K_0 + 2(15K_3 - 6K_3 + K_3) = 1$, so $K_3 = 1/20(1 - K_0)$. Similarly, $K_2 = -6/20(1 - K_0)$ and $K_1 = 15/20(1 - K_0)$.

   (b) First write R_0^2 in terms of K_0 so that

$$
\begin{aligned}
R_0^2 &= K_0^2 + 2(K_1^2 + K_2^2 + K_3^2) \\
&= K_0^2 + 2(1 - K_0)^2 \left[\left(\frac{15}{20}\right)^2 + \left(\frac{6}{20}\right)^2 + \left(\frac{1}{20}\right)^2 \right] \\
&= K_0^2 + 1.31(1 - K_0)^2.
\end{aligned}
$$

To minimise R_0^2, this should be differentiated with respect to K_0, set to zero, and solved for K_0. Differentiating, we get

$$\frac{dR_0^2}{dK_0} = 2K_0 - 2.62(1 - K_0).$$

Setting to zero and solving, we find that R_0^2 is minimised at $K_0 = \frac{131}{231}$ (which is a minimum since the second derivative is greater than zero for all K_0). Hence, the coefficients that minimise the expression are $K_0 = 131/231$, $K_1 = 75/231$, $K_2 = -30/231$ and $K_3 = 5/231$, with $K_{-1} = K_1$, $K_{-2} = K_2$ and $K_{-3} = K_3$.

Remark: the equation that was minimised in this question is the ratio of the variance of the errors of the filtered process to the variance of the errors of the original process, assuming that these errors are uncorrelated ($\text{Var}(e_t')/\ \text{Var}(e_t)$ using the notation of the notes). Thus th<e filter given above is the filter that is exact on 5-th degreee polynomials and that also minimises the ratio of the variances of these errors. This filter is sometimes called the 'Savitsky-Golay' filter.

□

Ex. 22 Let X be a process defined by $(1 - \phi B)X_t = \varepsilon_t$.

1. Compute its spectral density.
2. Compute the spectral density of $\eta = X_t - \frac{1}{\phi}X_{t-1}$.
3. What does that mean for the AR(1) process ?

Correction:

1. We use the spectral filtering theorem, and we can show that (note that we don't need to invert the polynomial actually as we will be multiplying complex numbers):

$$f_X(\omega) = \frac{\sigma^2}{2\pi} \frac{1}{|1 - \phi e^{-i\omega}|^2}.$$

2. We use the same result again:

$$f_\eta(\omega) = \frac{\sigma^2}{2\pi\phi^2} \frac{|\phi - e^{-2i\pi\omega}|^2}{|1 - \phi e^{-i\omega}|^2}.$$

3. If we multiply the denominator by $|e^{2i\pi}|^2 = 1$, we get $f_\eta(\omega) = Cte$, that is η is a white noise. In other words, we just have showed that an AR(1) process can be written in different ways.

□

Ex. 23 Let c be a real autocovariance function with finite number of non-null values

1. Show that the spectral density associated writes:

$$f(\omega) = \frac{1}{2\pi}\left(c_0 + 2\sum_{h=1}^{\infty} c_h cos(\omega h)\right).$$

2. Define a stationary process that has for spectral density:

$$f(\omega) = \frac{1}{2\pi}(5 - 4\cos(\omega))$$

Correction:

1. That's just using the symmetry of c and the definition of cosinus with complex numbers.
2. We define it as a moving average (which has finite number of non null covariances) of order 1, defined with adequate choice of parameters (solve the system of equations, with unknown noise and θ).

$\square$

Ex. 24 Let $Y_t = w_t + \theta w_{t-2}, \theta \neq 0, t \in \mathbb{Z}$, where $w \sim (0, \sigma^2)$ and $\sigma^2 > 0$.

1. Is the process an ARIMA process? If so which?
2. For which θ is the process invertible? Write it under the form:

$$w_t = \sum_{j=0}^{\infty} \pi_j Y_{t-j}.$$

3. Show that Y_t mighe be written as an infinite order autoregressive process.
4. Write the Box-Jenkins h-steps ahead forecast Y^n_{n+h}, $h = 1, 2, \ldots$, based on the infinite past $Y_n, Y_{n-1}, \ldots$. Write the variance of the forecast error.

Correction:

1. It's an ARIMA(0,0,2).
2. $(1 + \theta B^2)^{-1} = \sum_{i=0}^{\infty} \theta^i B^{2i}$. Hence the form required with $\pi_j = \mathbf{1}_{i \text{ even}}(-\theta)^{j/2}$.
3. Using the question above we can write $w_{t-2} = \sum_{j=0}^{\infty}(-\theta)^j Y_{t-2-2j}$, substituting in the initial definition of Y we get the required form.
4. Using the initial definition of the process we can write the Box-Jenkins forecast as:

$$Y^n_{n+h} = \begin{cases} \theta w_{n-1} & \text{if } h = 1 \\ \theta w_n & \text{if } h = 2 \\ 0 & \text{otherwise.} \end{cases}$$

The variance of the forecast error is then:

$$Var(Y_{n+1} - Y_{n+1}^n) = Var(w_{n+1} + \theta w_{n-1} - \theta w_{n-1}) = \sigma^2$$

and,

$$Var(Y_{n+2} - Y_{n+2}^n) = Var(w_{n+2} + \theta w_n - \theta w_n) = \sigma^2.$$

While for higher h's:

$$Var(X_{n+h}) = \sigma^2(1 + \theta^2).$$

□

Ex. 25 Let Y_t and η_t be weakly stationary independent time series with zero means and resp. spectral densities f_Y and f_η. Let D be a positive integer. Prove that the spectrum of the time series defined by:

$$X_t = Y_t + AY_{t-D} + \eta_t, \qquad A \in R$$

is given by

$$f_X(\omega) = [1 + A^2 + 2A\cos(\omega D)]f_Y(\omega) + f_\eta(\omega).$$

Correction: By independence:

$$\begin{aligned}\gamma_X(k) &= cov(Y_t + AY_{t-D} + \eta_t, Y_{t-k} + AY_{t-D-k} + \eta_{t-k}) \\ &= \gamma_Y(k) + A\gamma_Y(k-D) + A\gamma_Y(k+D) + A^2\gamma_Y(k) + \gamma_\eta(k).\end{aligned}$$

This leads to the following spectrum, using the definition of the notes. After computation we get the formula required.

□